What is central banks'mandate ? A study of monetary policy through its communication starting 2000, focusing on the ECB, the Fed and the BoE

Anonymous Author(s)
Affiliation
Address
email

1 Introduction

After a long time of focus on fighting inflation and, in the United States, on supporting economic activity and employment, monetary policy has seen its scope of responsibility steadily expand. For instance, the 2008 crisis cemented its mandate for macroprudential regulation of the financial system. Together with the COVID-19 crisis, it further extended this scope, with central banks stepping in to preserve sovereign debt sustainability through unprecedented asset purchase programmes. Today, a new frontier is emerging: whether action against climate change falls within the purview of central banks — a debate that divides the profession, with proponents arguing that climate risk is a financial stability concern (Schnabel, 2021) and critics warning against mandate overreach. That said, it would be interesting to assess the history of this monetary policy expansion in detail. How have central banks communicated about these evolving responsibilities over time ? And does their communication reflect the same institutional priorities, or do the ECB, the Federal Reserve and the Bank of England speak with fundamentally different voices ? In this dissertation I would like to revisit this history by examining how the various issues central banks have faced over time are reflected in their communication. Indeed, communication has become an integral part of monetary policy. Since the early 2000s, central banks have increasingly relied on forward guidance — the practice of signaling future policy intentions to anchor market expectations. This signaling effect, now widely recognized as a transmission channel in its own right, means that what central bankers say matters as much as what they do. Thus, analyzing central bank speeches offers a privileged window onto the evolution of these expanding mandates.

In economics, numerous studies focus on central bank communication. They usually pursue two distinct but complementary objectives. A first strand of literature seeks to measure the causal impact of central bank communication on financial markets and macroeconomic outcomes. A foundational contribution is Blinder (2008), who survey the theory and practice of central bank communication and document how transparency has become a policy instrument in its own right. In this literature, it is common to rely on NLP tasks, after all text is data (as Gentzkow et al., 2019 put it). For example, Hansen & McMahon (2016) exploit Fed minutes to causally identify the market effects of FOMC communication, distinguishing between the information and the Delphic components of forward guidance. A second strand, closer to my approach, seeks instead to document and historicise the thematic content of central bank communication — asking not what effect it has, but what central banks actually talk about and how this has evolved. For instance, Ferrara et al. (2022) employ structural topic modeling on the Monetary Dialogues between Members of the European Parliament and the ECB President from 1999 to 2019, studying how elected officials hold central bankers accountable. More recently, Campiglio and al. (2025) build a large dataset of central bank speeches worldwide and study the emergence of climate-related communication, relying on a dictionary approach to identify relevant passages.

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My dissertation belongs to this second strand, but departs from existing contributions in two respects. First, unlike dictionary-based approaches such as Campiglio and al. (2025), I apply BERTopic (see Grootendorst (2022)), a neural topic modeling approach based on contextual embeddings), which allows topics to emerge inductively from the data rather than being imposed by a pre-defined vocabulary. Second, I conduct a systematic three-way institutional comparison between the ECB, the Federal Reserve and the Bank of England over the 2001–2023 period, covering the full arc from the pre-crisis consensus to the post-COVID inflation surge.

2 Description of the data

2.1 The Campiglio et al. (2025) Dataset

The data I will be using is a corpus of central banker speeches assembled by Campiglio, Deyris, Romelli, and Scalisi for their article published in 2025 in the European Economic Review.
The dataset comprises 35,487 unique speeches from 131 central banks, for the period January 1986 — date of the first online speech by a central bank —to December 2023. Among these, 10,544 originate from the three institutions that I study here: the Federal Reserve, the Bank of England and the European Central Bank. The authors collected speeches from three complementary sources : (i) the Bank for International Settlements (BIS); (ii) the official websites of individual central banks; and (iii) central bank archives for older documents. Speeches originally delivered in languages other than English were translated using Microsoft Translator, a multilingual machine translation cloud service. The dataset is further enriched with original metadata coding the gender and institutional role of each speaker — categorized as governor, deputy governor, board member, or senior management — based on publicly available information.

2.2 Sample Selection

For my study, I restrict the analysis to speeches from three institutions: the Bank of England (BoE), the Board of Governors of the Federal Reserve,[1] and the European Central Bank (ECB). This selection yields a balanced and comparable corpus across three major currency areas — the United Kingdom, the United States, and the euro area — that together account for a large share of global monetary policy communication.

Within these institutions, I further restrict the sample to speeches delivered by governors and deputy governors only. This choice is motivated by two considerations. First, these positions are strictly comparable across institutions: in all three cases, the governor is the highest-ranking official setting monetary policy, while deputy governors form the inner circle of decision-making. By contrast, the category of "board member" covers very different realities depending on the institution — Federal Reserve Governors are presidential appointees with a national mandate, while Bank of England Monetary Policy Committee members include external academics whose speeches may resemble research papers more than policy communication.[2] Second, restricting to governors and deputy governors ensures that the corpus reflects official institutional communication rather than individual academic contributions. This leaves a final sample of 2,286 speeches, distributed as follows: 473 from the Bank of England, 820 from the Federal Reserve, and 993 from the ECB, covering the period 2001–2023.

[1] I exclude the 12 Federal Reserve Banks to focus exclusively on the institution setting monetary policy at the national level, and to maintain comparability across the three central banks.

[2] A notable example is the speech by Stephen Nickell, MPC member and LSE professor, which contains a full academic bibliography and statistical tables — a format incompatible with the other institutions' communication style.

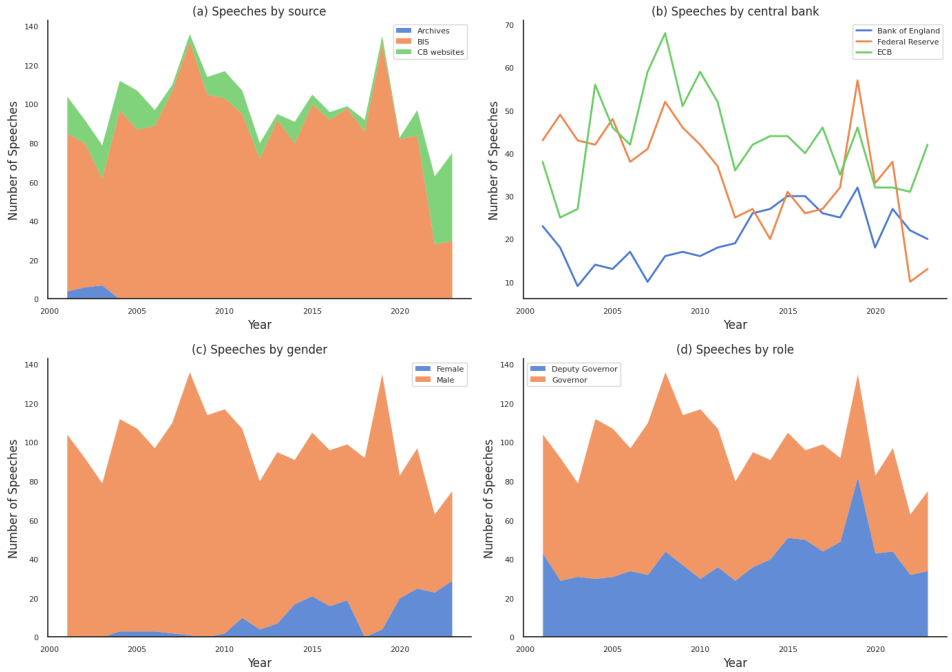

Figure 1: Central bank speeches over time, 1921–2023; *Scope: Speeches by Governors and Deputy governors of the Boards of Governors of the Federal Reserve, the ECB, and the Bank of England, as compiled by Campiglio et al.*

2.3 Text Preprocessing

The dataset needed some preprocessing for the NLP tasks. I relied on the rich metadata to do so, applying institution-specific cleaning rules where appropriate. I identified and removed several categories of artifacts. For istance, standard footers appended by the ECB to all its online publications — containing contact information and copyright notices — were systematically stripped. Inline footnote markers of the form `[1]`, `[2]`, etc., which appeared embedded within sentences as citation references, were removed throughout the corpus. Third, page number artifacts introduced during PDF-to-text conversion — typically appearing as patterns such as `1/2001 5` — were deleted. Fourth, encoding artifacts produced by imperfect character recognition during PDF extraction, such as `Rf6ntgen` for *Röntgen* or `He4me4le4inen`, were corrected. Fifth, missing apostrophes in Federal Reserve speeches — a systematic artifact of PDF conversion affecting contractions such as *Id*, *Its*, *Whats* — were restored using targeted substitution rules. For speeches containing academic reference sections — a format found almost exclusively among Bank of England board members, and therefore absent from the final sample — bibliographies were removed by detecting section headers such as *References* or *Bibliography* appearing in the final quarter of the document, while carefully preserving in-text citations such as *"the references therein"* or *"as references show"*. [3]

2.4 Chunking Strategy

As I want to do some topic modelling using BERTopic, I need my corpus to be homogeneous. The fact of keeping only Governor and Deputy Governors' speeches should have ensured this. Now, I have to worry about text length and heterogeneity in text length. Indeed, I seek to analyse the different topics that emerge in central bankers'speech. However usually, a central banker refers to various topics in a same speech.

Hence, the information I seek is available at a paragraph level. This is one motivation for the division of each text into chunks. Another motivation is that I apply a transformer-based embedding model which has a context window of 512 tokens. The advantage of this embedding model —

[3] For this processing part, I relied on the LLM claude to detect all possible problems. I used it too to write this paragraph, to be as more precise as possible on the preprocessing step.

`hugom123/bge-centralbank` — is that it was specifically fine-tuned on central bank communication. Thus, it might deliver more precise results in the topic modelling analysis than more general models. However, the average speech in the sample contains approximately 3,200 words, far exceeding this limit. I need to chunk my database, otherwise all speeches exceeding 512 tokens will be truncated, thus I would loose potentially valuable information.

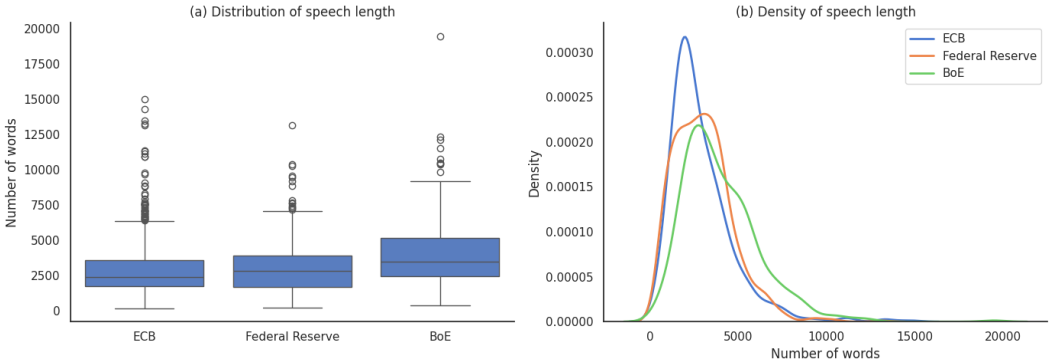

Figure 2: Speech length by central bank — Governors & Deputy Governors, 2001–2023 *Metric : number of words; Scope: speeches by members of the Boards of Governors of the Federal Reserve, the ECB, and the Bank of England, as compiled by Campiglio et al.*

To address this, I adopt a sentence-based chunking strategy. Each speech is first segmented into sentences using spaCy's dependency parser ($en_core_web_sm$), which correctly handles abbreviations and numerical expressions common in financial texts — such as *e.g.*, *U.S.*, or *2.5%* — that would cause errors with naive period-based splitting. Sentences are then aggregated into chunks of at most 450 tokens, with an overlap of two sentences between consecutive chunks to preserve contextual continuity at boundaries. This procedure yields 4,941 chunks from 483 sampled speeches, with an average chunk size of 343 words and a standard deviation of 58 words, indicating a highly homogeneous chunking. One limitation of this approach is that the natural paragraph structure of the original speeches is not preserved, as the corpus was not available with paragraph-level markup. Furthermore, some thematic segments may be longer or shorter than the 450-token threshold, meaning that a single topic may occasionally be split across two consecutive chunks, or conversely that two distinct topics may be merged into one.

3 Presentation of the model

3.1 The choice of model

The goal of my project is to find a way to discover the latent thematic structures of my corpus, knowing that I do not have any prior labeling or categorization of my speeches. Indeed, I could technically create labels on a small part of my corpus and then train a model on this hand-made categorization, validate the model and apply it on the whole set. But this would be very costly in time. It could be considered though as an extension to do someday to challenge these BERTopic results. As I cannot do this, I rely on topic modeling, a ML technique which consists in uncovering the latent topics of a corpus unsupervisedly.

There are many topic modeling algorithms. The classical ones rely on the bag-of-words paradigm. For example, in the classic Latent Dirichlet Allocation model, each document is represented as a distribution over topics, and each topic as a distribution over words — but words are treated as independent and interchangeable tokens, stripped of their syntactic and semantic context. For example, the word *easing* has different meanings depending on the context in central banker's speeches :

- In the sentence *The committee decided on further easing of monetary policy*, it refers to the quantitative easing policy, that is massive asset purchases by central banks that were introduced in the aftermath of the 2008 crisis.

- In *The easing of supply chain tensions contributed to lower inflation*, the word refers to supply-side normalisation.
- Last, consider *The easing of regulatory requirements for small banks*, here the word easing refers to financial deregulation.

Thus, the word *easing* does refer to many different economic realities and policies to be implemented. That is why, the analysis would gain to be conducted by models that account for semantic richness. That is what BERTopic does(Grootendorst2022). In the following section, I will describe its functioning.

3.2 BERTopic

BERTopic relies on three major steps.

Embedding First, it necessitates representing each document as a dense vector in a high-dimensional semantic space. That is building embeddings for each documents. For the first step of BERTopic, as said in the previous part, I decide to rely on a sentence transformer that produces embeddings through the self-attention mechanism, and thus, takes context into account. I chose *hugom123/bge-centralbank*, a free-to-use model based on the *BAAI/bge-large-en-v1.5* architecture and fine-tuned on central bank communication.

Cluster building The previous step provides me with 1,024-dimensional chunk embeddings. That is massive. To overcome the curse of dimensionality, BERTopic uses the UMAP model to reduce dimensionality. In his article, Grootendorst2022 explains that UMAP enables to preserve both local and global features of high-dimensional data more faithfully than alternatives such as PCA or t-SNE. Then, it builds clusters based on HDBSCAN.

Topic representation Last, once clusters have been identified, BERTopic represents each of them using a variant of TF-IDF called c-TF-IDF. Rather than computing term frequencies at the document level, c-TF-IDF computes frequencies through each clusters, treating all chunks belonging to a cluster as a single meta-document. The advantage of c-TF-IDF is that it is robust to time.

3.3 Fine-tuning of the model

Hyperparameters choice The quality of BERTopic results is sensitive to the choice of hyperparameters, particularly those governing the UMAP dimensionality reduction and the HDBSCAN clustering. A key parameter is `n_neighbors` in UMAP, which controls the balance between local and global structure in the reduced embedding space: a small value preserves fine-grained local relationships between chunks, while a larger value captures broader semantic neighborhoods. After experimenting with several configurations on a stratified sample of my whole dataset [4], I set `n_neighbors` =50 , `min_dist` =0.1, and `min_cluster_size` =50 in HDBSCAN, which produced 109 interpretable topics on the full corpus of 30,401 chunks.

Outliers One limitation of HDBSCAN is that it assigns documents lying in low-density regions of the embedding space to a noise cluster, labelled Topic 1. In my first results, this outlier cluster contained 10,578 chunks, representing 34.8% of the corpus — a share too large to be ignored. To address this, I apply the `reduce_outliers` method with the *embedding* strategy: each outlier chunk is reassigned to the topic whose centroid is closest in the original 1,024-dimensional embedding space, using cosine similarity. This strategy is useful as it prevents me from rerunning the model which is costly in energy. After outlier reduction, all 30,401 chunks are assigned to a substantive topic.

Topic Aggregation BERTopic produced 109 topics after outlier reduction. This is a lot, so I decided to merge some topics. I proceeded manually.

1. First, I computed pairwise cosine similarities between topic embeddings — the weighted average of chunk embeddings belonging to each topic — and merged topics whose similarity exceeded 0.7, indicating strong semantic proximity.

[4] I computed the model on a subset first to use less energy, although I had the chance to use onyxia's gpu.

2. Second, I inspected the top 10 most representative words of each topic and its representative documents to verify that the merge was economically meaningful.

This granular inspection of topic pairs proved valuable beyond the immediate aggregation decision. By systematically examining what BERTopic had grouped together — and what it had kept apart — I gained a deeper understanding of the semantic structure the model had identified in the corpus. Several findings emerged naturally from this process: the model correctly distinguished between the *implementation* of Quantitative Easing (Topics 29, 42, 73), its *market consequences* in terms of yield compression (Topic 57), and its *communication dimension* through forward guidance (Topic 74). Similarly, it separated the ECB's institutional language around European fiscal rules from the Fed's more outcome-oriented fiscal discourse — a distinction that would have been lost under a purely algorithmic aggregation.

After aggregation, the final taxonomy comprises **43 topics**, which I interpret and name based on their top words and representative documents in the results section.

4 Results

4.1 Heatmap of central banks'communication

To begin with, let us see what our three central banks talk mostly about in the whole corpus.

The heatmap shows that, as one would have guessed, macroeconomic outlook is the dominant topic for all three institutions (BoE: 16.2%, Fed: 10.3%, ECB: 7.1%), confirming that the assessment of current economic conditions remains the backbone of central bank communication regardless of institutional context. However, BERTopic manages to retrieve the most important institutional differences between central banks. Several patterns stand out immediately.

- First, the ECB corpus is dominated by topics that reflect its unique institutional position within the eurozone : European Integration (15.6%) and European Governance (10.8%) together account for more than a quarter of ECB speeches — topics that are virtually absent from Fed (0.2%) and BoE (0.7% and 0.3%) corpora. This reflects the ECB's dual role as both a monetary authority and a key actor in the construction of European economic governance.
- As for the Fed, its speeches reflect its broader social mandate and domestic orientation : Labour Market (7.2%), Housing & Mortgages (5.4%), Crisis Liquidity (7.7%), and Social Outreach (3.0%) all rank higher for the Fed than for the other two institutions.
- The BoE occupies an intermediate position, with a relatively balanced profile across topics, but stands out on Banking Regulation (12.7%), Resolution & Bail-in (5.3%), and Digital Finance & Crypto (5.5%). The prominence of resolution topics reflects the BoE's post-2008 focus on ending "too big to fail" — a priority championed by Mervyn King and Paul Tucker, while the high share of Digital Finance reflects the BoE's early engagement with fintech regulation and CBDC research, driven in particular by Jon Cunliffe.

One interesting remark is that even at this stage we can see the difference of global mandate between central banks. Indeed, whereas the Fed speaks about Labour market as a direct policy objective, the ECB addresses labour almost exclusively through the lens of Structural Reforms (7.9%). If we take a closer look and consider the TF-IDF for each topic in the specific chunks of each central banks, we can clearly see the mandate gap on labour market between both central banks. For the **Fed**, the dominant terms are *productivity*, *labor*, *unemployment*, *employment*, *job*, *workers*, *participation*, and *labor force* — an operational vocabulary oriented toward measuring and closing labour market gaps. The BoE mirrors this profile closely, with *labour*, *productivity*, *unemployment*, *pay*, *wage*, and *demand* all featuring prominently, consistent with its inflation-targeting mandate in which wage dynamics are a direct transmission channel. Both institutions speak about the labour market as something they are expected to *diagnose and respond to*. The ECB's lexicon tells a different story. The most common words it uses for the Structural Reforms topic are *labour*, *reforms*, *countries*, *structural*, *competitiveness*, *employment*, *structural reforms*, *states*, and *europe*. The word *states* is telling: it signals that the ECB is not discussing labour markets as a policy lever it controls, but as a domain it addresses *through* member states.

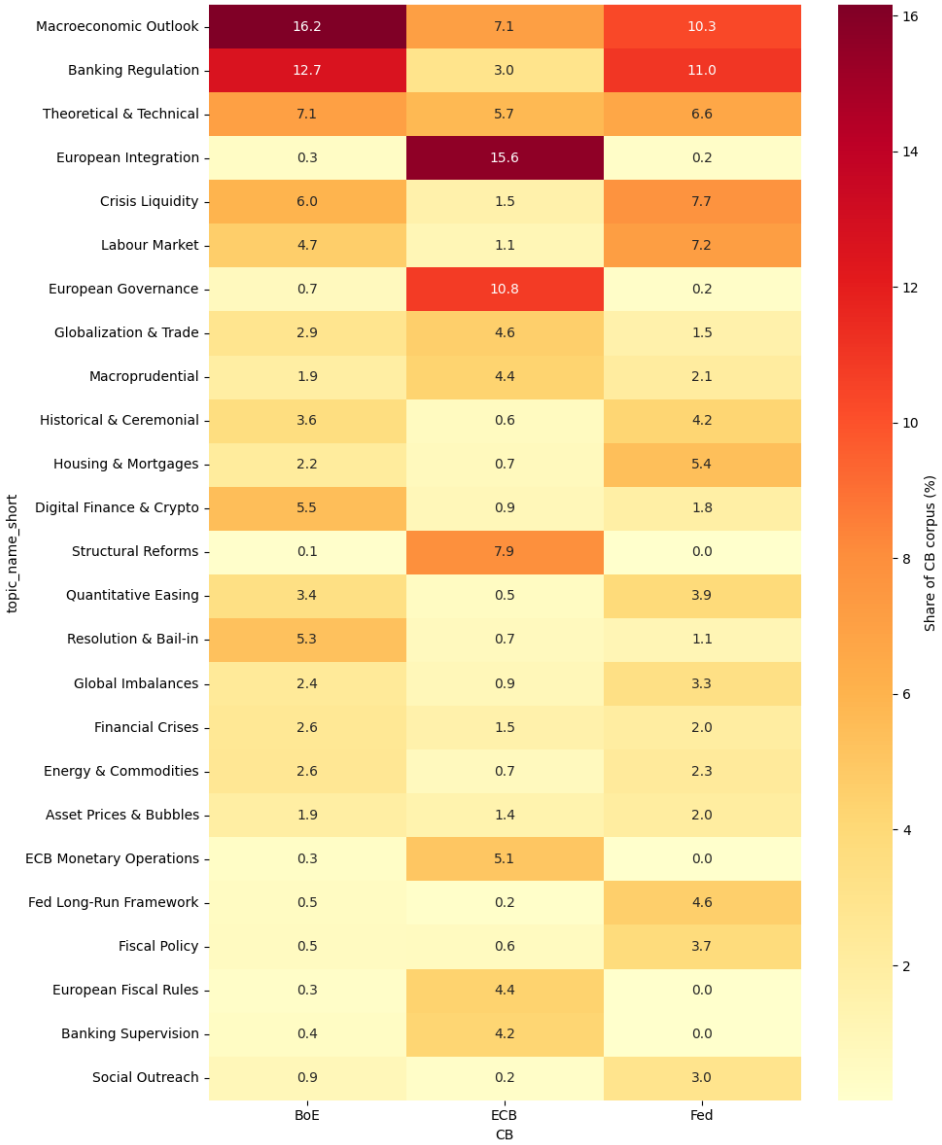

Figure 3: Topic Distribution by Central Bank(% of each CB corpus)

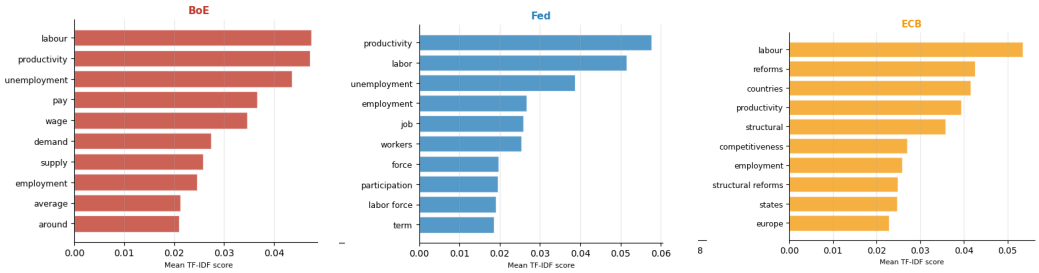

Figure 4: Top TF-IDF terms by central bank for the *Labour Market* topic (BoE and Fed) and the *Structural Reforms* topic (ECB).

4.2 Evolution of key topics

Let us now look at the evolution of monetary policy in more detail, assessing the big crisis and the new policy tools that followed.

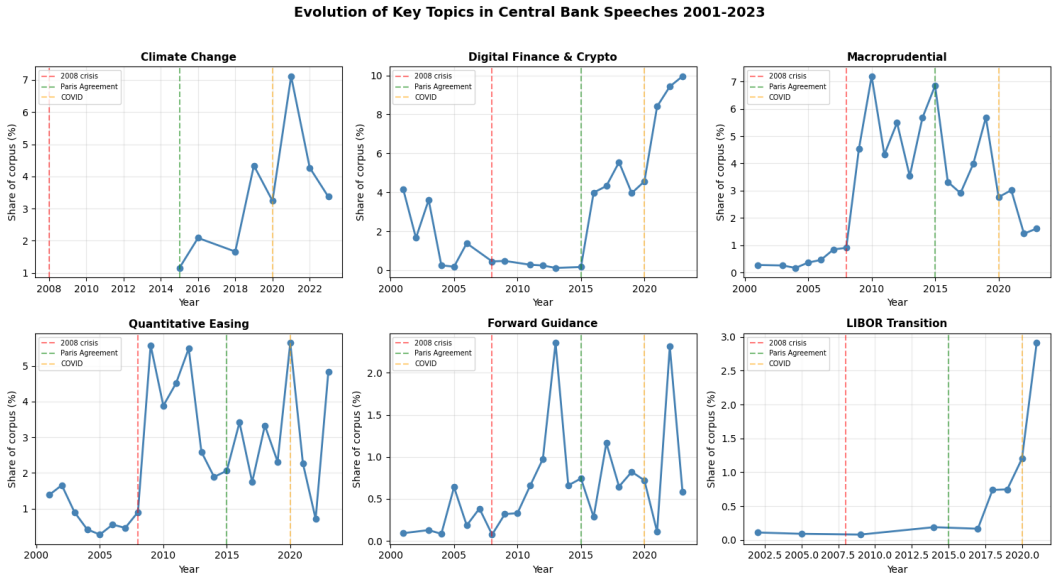

Figure 5

This figure documents the temporal evolution of six topics that have emerged or transformed significantly over the 2001–2023 period.

- Digital Finance & Crypto exhibits a two-phase pattern. A first wave of attention in 2001–2003 relates to early e-payments and electronic money regulation. A long trough follows until 2015, when a second and far more intense wave begins, driven by the rise of Bitcoin, fintech disruption, and the CBDC debate — culminating in a peak around 2022 that reflects regulatory urgency around stablecoins and crypto assets.

- Macroprudential policy shows a sharp and durable increase after 2008, peaking around 2009–2010 as central banks articulated the lessons of the financial crisis. The subsequent volatility reflects the iterative construction of macroprudential frameworks — Basel III, stress testing, countercyclical buffers — which generated sustained policy debate throughout the 2010s.

- Quantitative Easing peaks immediately after 2008 (Fed and BoE) and again after 2015 (ECB), before a third surge in 2020 (COVID). Forward Guidance shows a similar post-2008 emergence, with a notable peak around 2013–2014 reflecting the Fed's explicit threshold guidance.

- Finally, climate Change shows a clear structural break after 2015, consistent with the Paris Agreement marking a turning point in the awareness of climate risk among financial institutions. The sharp peak in 2020–2021 reflects the convergence of three forces: the COVID-19 crisis, which prompted green recovery narratives; Christine Lagarde's arrival at the ECB with an explicit green agenda; and the intensification of the Network for Greening the Financial System (NGFS) work programme.

4.3 More difference between central banks, by topics. Example of climate change and Quantitative easing

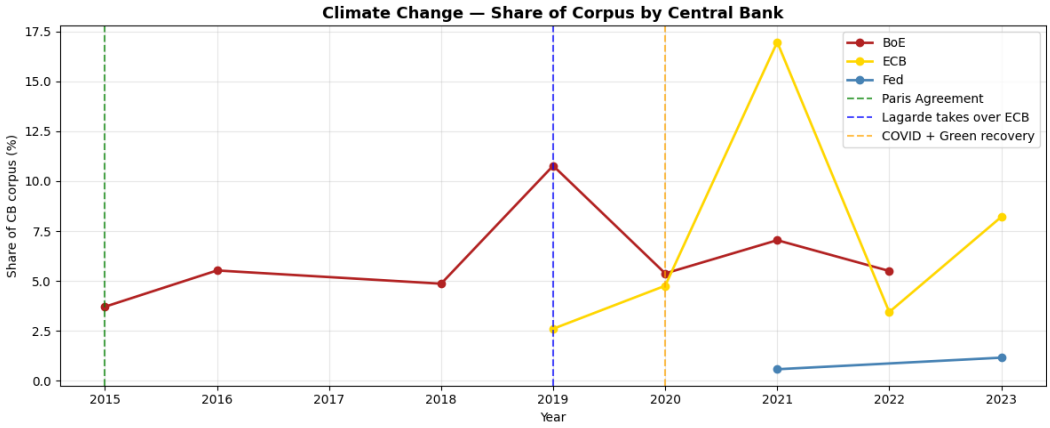

Figure 6

The strikingly low share of climate communication in the Fed corpus (below 1% throughout the period) contrasts sharply with the ECB (peak of 17% in 2021) and the BoE (sustained above 5% since 2016). This reflects a genuine institutional difference in mandate interpretation. The ECB under Lagarde has explicitly integrated climate change into its monetary policy strategy review (2021), arguing that climate risk is a financial stability concern that falls within its mandate. The BoE, under Mark Carney from 2013, was a pioneer — Carney's 2015 Lloyd's of London speech on the "tragedy of the horizon" was a landmark moment in central bank climate engagement. The Fed, by contrast, has remained much more cautious, reflecting both political constraints (Republican opposition to "mission creep") and a narrower interpretation of its mandate. Jerome Powell has repeatedly stated that climate policy is Congress's prerogative, limiting Fed governors' public engagement on the topic. [5]

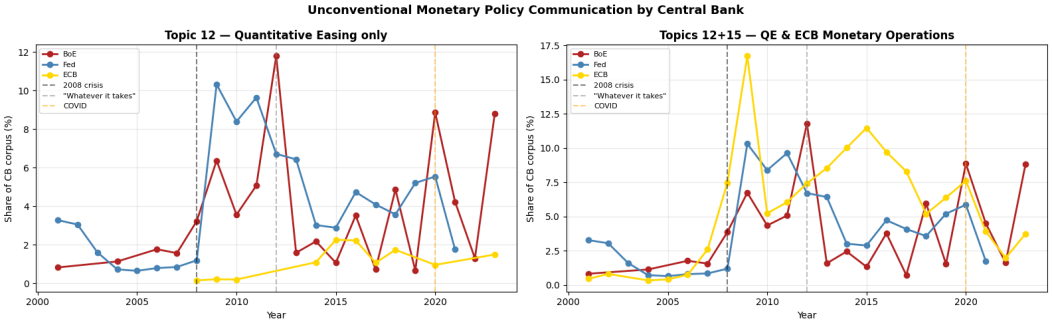

Figure 7

This figure illustrates the temporal evolution of unconventional monetary policy communication across the three central banks, distinguishing between the QE topic strictly defined (Topic 12, left panel) and a broader measure that combines QE with ECB-specific monetary operations (Topics 12+15, right panel).

Once ECB-specific monetary operations (Topic 15) are added, the ECB curve shifts dramatically upward, with a peak as early as 2008–2009. This suggests that the ECB communicated extensively about liquidity provision and monetary operations from the outset of the crisis, but through a distinct institutional vocabulary rather than the asset purchase terminology dominant in Anglo-Saxon

[5] "The Federal Reserve is not and will not be a 'climate policymaker'." See *https://www.federalreserve.gov/newsevents/pressreleases/powell-statement-20231024b.htm*

discourse. A simple corpus inspection confirms this: among ECB chunks assigned to Topic 15 over the 2008–2012 period, 32.9% explicitly contain the phrase *non-standard measures* — the ECB's preferred framing for its crisis interventions, which deliberately avoided the term *quantitative easing*.

5 Conclusion

I analyzed the underlying themes in the official statements of the governors and deputy governors of the Fed, the BoE, and the ECB. This revealed both similarities and differences that reflect what we know about their respective mandates (the ECB is less focused on reducing the unemployment gap than the other central banks). One finding is the difference in stance on climate issues. Furthermore, the exploratory analysis shows that a more detailed study of the topics would allow for a better understanding of the differences in the lexical fields used by each institution. For instance, regarding quantitative easing, the Fed and the BoE on one side and the ECB on the other do not seem to use the same terminology. It might be interesting to explore this further in a subsequent analysis. Finally, in terms of validity, the topic aggregation stage—which involved comparing the cosine similarity of closely related topics as well as their recurring words—showed that, with a few exceptions, the topics appear fairly consistent. However, this could later be compared to LDA results or even to manual topic modeling.

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References

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